

Modeling CIN Cancer Evolution: A Dual Population Approach for Astrocytoma



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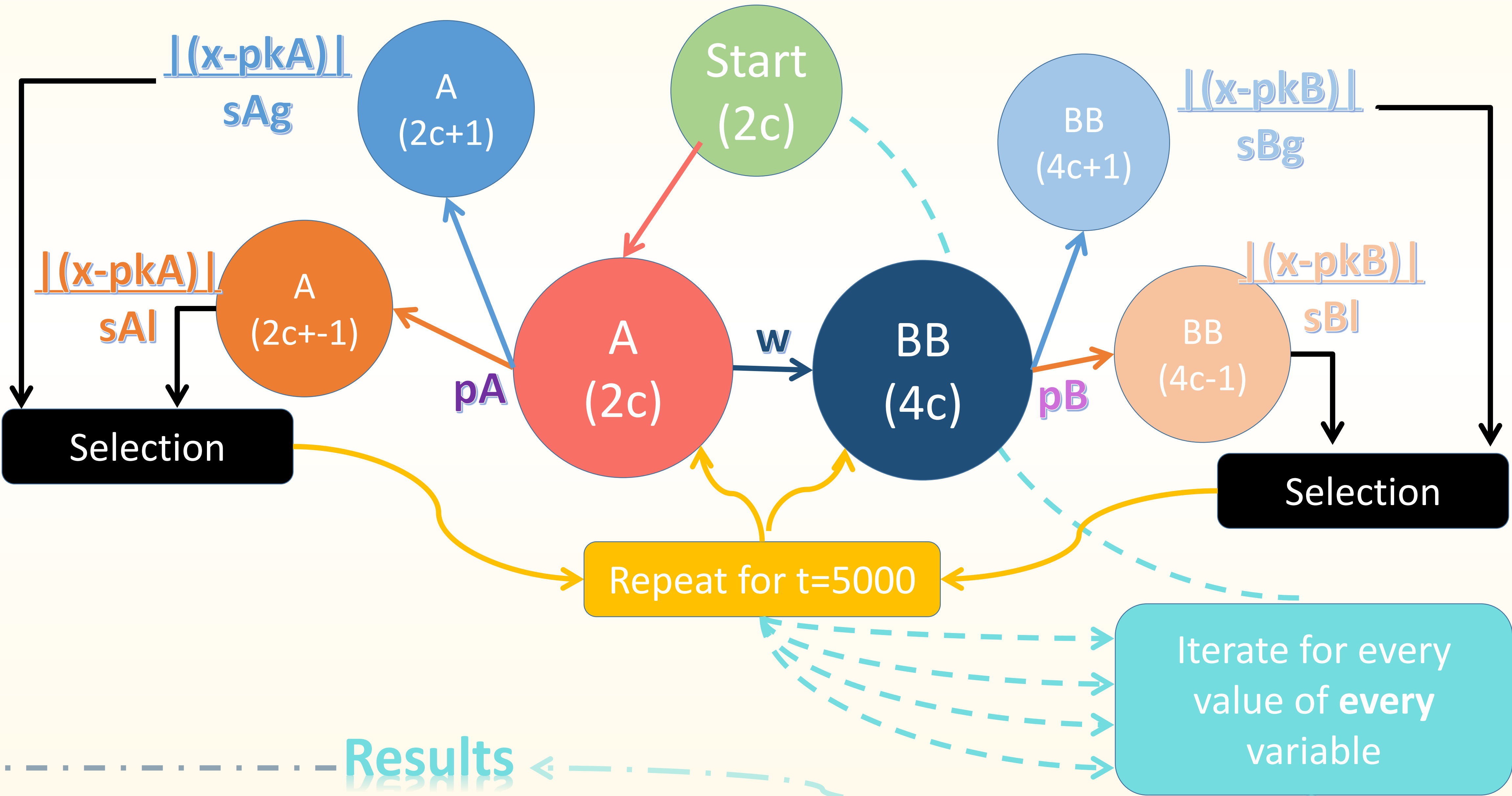
¹University of Guelph, ²Fractal Therapeutics, ³Notable Contribution

Background

Methods

- Chromosome Instability (CIN)** in cancer cells leads to abnormal karyotypes, wherein chromosomes are duplicated, lost, fragmented, or blended together.
- One trend: Bimodality.** The same type of cancer can have very different chromosome counts, having a peak close to standard chromosome content (2c), and a peak at a highly increased counts (3c, 4c, etc.), but fewer intermediate counts between those peaks.
- Explanation: Whole Genome Doubling (WGD),** where a mitotic mishap makes it double the existing count.
- Problem:** The high count peak often doesn't match what we'd expect from WGD, and has a very different shape in count distribution from the 2c peak.

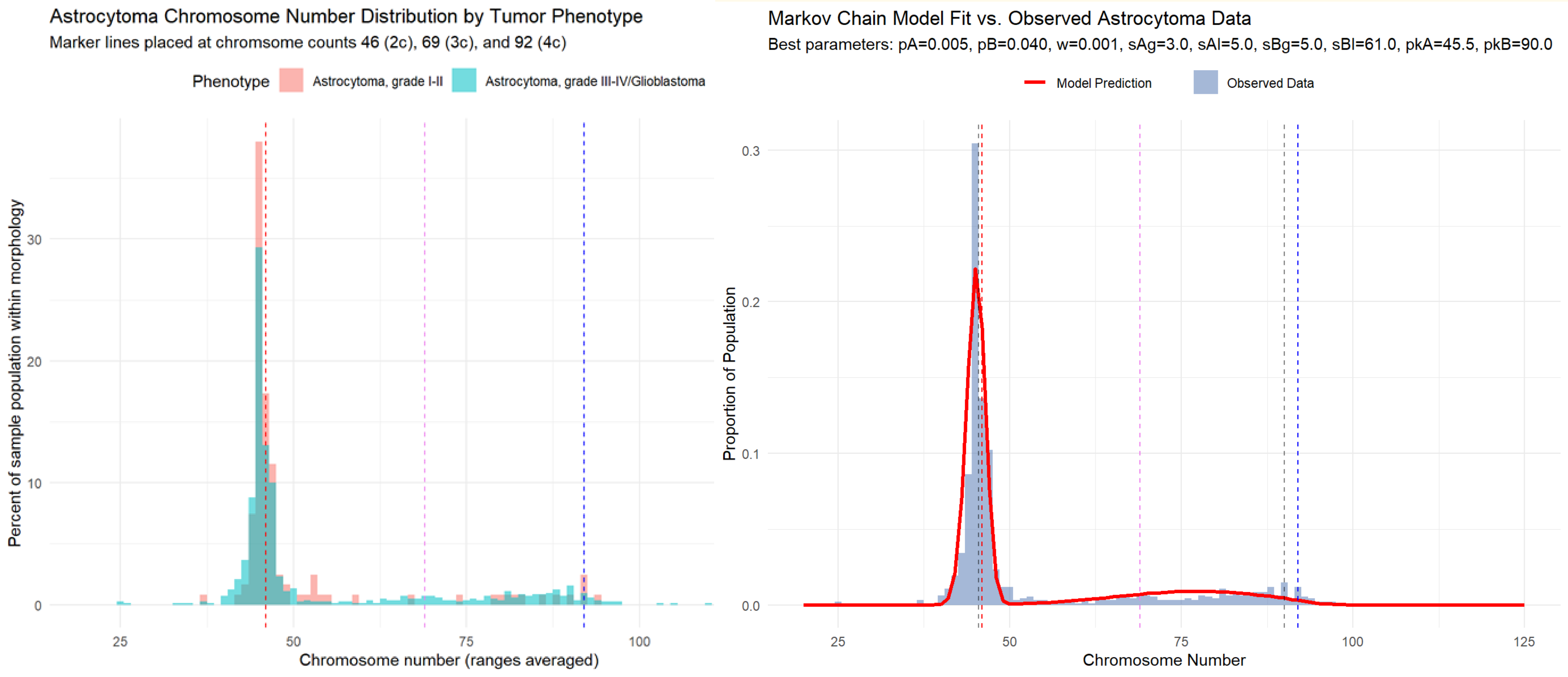
- Possible solution:** Two cancer populations, using different evolutionary strategies.
- To test, we use a **Markov Chain**, followed by a **grid search**.



Discussion

- Wide distribution after simulated WGD** indicates high missegregation rate, differing from original population.
- Two strategies:** Conserve starting chromosome number with low variance, or pursue new options with high variance.
- Further investigation:** Pathology link, other cancer types, poly-populations?

Results



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